Flashing the Jetson Bootloader using L4T

1. Downloading Files

Open the following link in your web browser and select Jetson Linux version 38.4.

```
https://developer.nvidia.com/embedded/jetson-linux-archive
```

Jetson Linux 38.x

JETSON LINUX VERSION	Jetson T5000	Jetson T4000	Jetson AGX Orin	Jetson Orin NX	Jetson Orin Nano
38.4 > January 2026	✓	✓			
38.2/38.2.1 > August 2025	✓				

Click on [Driver Package (BSP)] to download the BSP file.

Jetson Linux 38.4 (1/06/2026)

Drivers	Driver Package (BSP)
	Camera SIPL
	Sample Root Filesystem
	Jetson Linux API Reference
Sources	Driver Package (BSP) Sources
	Sample Root Filesystem Sources

Jetson_Linux_R38.4.0_aarch64.tbz2

Download the BSP file to your Ubuntu computer or Ubuntu virtual machine. For this example, we will assume the file is downloaded to the `~/jetpack` folder.

2. Extracting Files

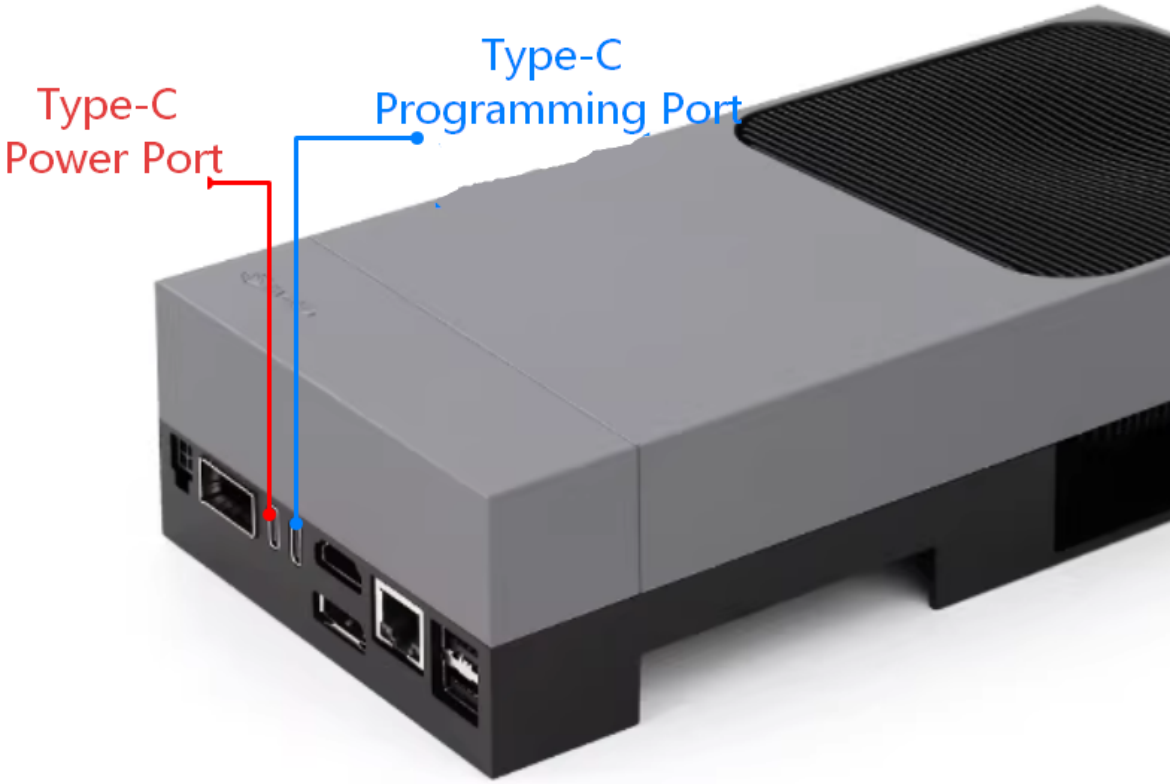
Open the terminal on your Ubuntu computer or virtual machine and run the following commands:

```
cd ~/jetpack
tar xvf Jetson_Linux_R38.4.0_aarch64.tbz2
```

```
yahboom1:~/jetpack$ tar xvf Jetson_Linux_R38.4.0_aarch64.tbz2
Linux_for_Tegra/
Linux_for_Tegra/p3737-0000-p3701-0008.conf
Linux_for_Tegra/jetson-orin-nano-devkit.conf
Linux_for_Tegra/jetson-agx-orin-devkit-industrial.conf
Linux_for_Tegra/igx-orin-devkit-qspi.conf
Linux_for_Tegra/jetson-agx-orin-devkit-as-nx-8gb.conf
Linux_for_Tegra/p3834-0008-p3740-xxxx-nvme.conf
Linux_for_Tegra/kernel/
Linux_for_Tegra/kernel/kernel_display_supplements.tbz2.list
Linux_for_Tegra/kernel/LICENSE
Linux_for_Tegra/kernel/SOURCE.kernel
```

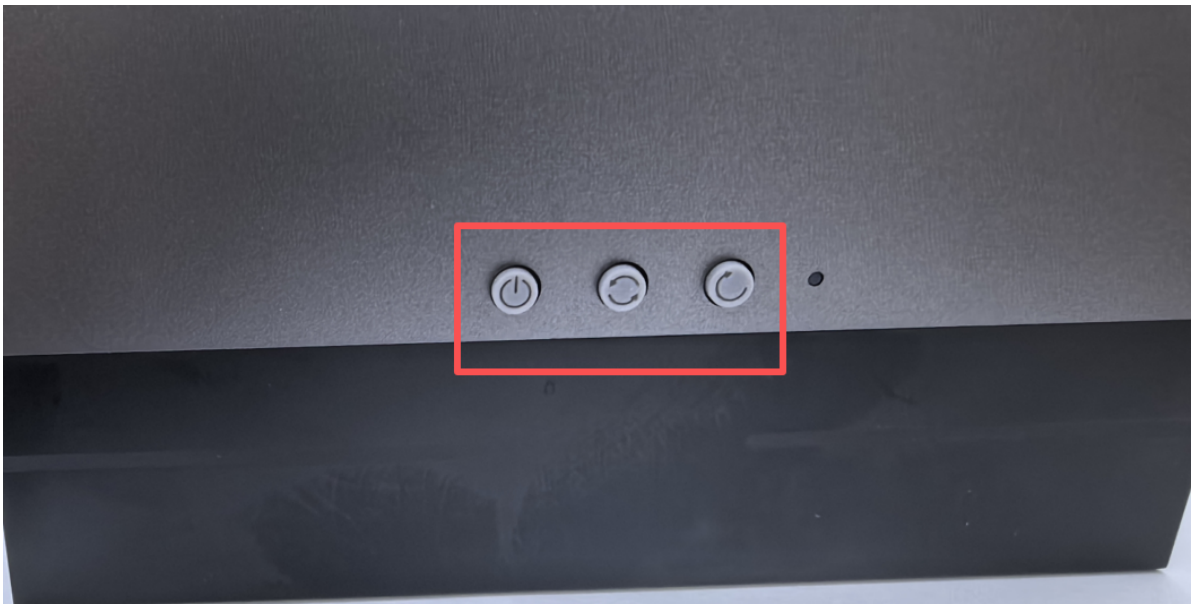
3. Hardware Connections

The Jetson AGX Thor mainboard requires a connection to both a power adapter and a Type-C data cable. The Type-C cable serves as the data link; one end must be connected to a USB port on your computer, while the other end must be connected to the specific Type-C port located right next to the HDMI port on the Jetson AGX Thor mainboard.



Connect the power adapter to the Jetson AGX Thor mainboard. Locate the three buttons at the rear of the board: press and hold the [Flash] button, then briefly press the [Reset] button. Wait for 3 seconds before releasing the [Flash] button to enter Flash Mode.

As shown in the figure below, the three buttons—from left to right—are the Power button, the Flash button, and the Reset button.



Open a terminal and check for connected USB devices. The appearance of `ID 0955:7026 NVIDIA Corp. APX` indicates that the connection has been successfully established.

```
lsusb
```

```
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 006: ID 0955:7026 NVIDIA Corp. APX
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
```

4. Start Flashing

Navigate to the JetPack system directory:

```
cd ~/jetpack/Linux_for_Tegra
```

Run the following command to begin flashing the bootloader. The entire process takes approximately 5 minutes; during this time, there may be extended periods where no output appears in the terminal, so please wait patiently.

```
sudo ./l4t_initrd_flash.sh --qspi-only jetson-agx-thor-devkit internal
```

```
yahboom1:~/jetpack/Linux_for_Tegra$ sudo ./l4t_initrd_flash.sh --qspi-only
jetson-agx-thor-devkit internal
/home/ayue/jetpack/Linux_for_Tegra/tools/kernel_flash/l4t_initrd_flash_internal.
sh --no-flash --qspi-only jetson-agx-thor-devkit internal
***********************************
*                                 *
*  Step 1: Generate flash packages *
*                                 *
***********************************
Run image gen script: /home/ayue/jetpack/Linux_for_Tegra/tools/kernel_flash/l4t_
create_images_for_kernel_flash.sh --external-device nvme0n1p1 -c tools/kernel_fl
ash/flash_l4t_t264_nvme.xml --qspi-only --boot-chain-flash ALL --boot-chain-sele
ct A jetson-agx-thor-devkit internal
Create folder to store images to flash
Generate image for internal storage devices
Generate images to be flashed
```

Once the message `Flashing finish` appears in the terminal output, the flashing process has completed successfully.

```
[flash_bsp_jetson-t264_die0]: [flashFirmware(343)] :      in flashFirmware
[flash_bsp_jetson-t264_die0]: [flashFirmware(354)] :  This is a l4t Run, Skippin
g FW flashing
[flash_bsp_jetson-t264]: start time  = 1775140302.662266
[flash_bsp_jetson-t264]: end time  = 1775140577.6197844
[flash_bsp_jetson-t264]: Total Time = 274.9575183391571
[flash_bsp_jetson-t264]: Image Flashing took 274.9575183391571
[flash_bsp_jetson-t264]: Flashing finished Successfully!!
/home/ayue/jp71/Linux_for_Tegra
Flashing finish
Cleaning up...
Log is saved to Linux_for_Tegra/initrdlog/flash_3-2_0_20260402-223142.log
```

The device will not power on automatically after the flashing process is complete. You must connect a display monitor and then power cycle the device to boot it up; you will then be able to see the boot information for version R38.4.0.